SIMATS ENGINEERING

Engineer to Excel

SIMATS

Saveetha Institute of Medical And Technical Sciences

(Declared as Deemed to be University under Section 3 of UGC Act 1956)

CO1 - ASSESSMENT TOOL 2

COURSE NAME: DESIGN AND ANALYSIS OF ALGORITHMS

COURSE CODE: CSA06

COURSE OUTCOME COVERED

CO1: Determine algorithm efficiency using asymptotic notations and mathematical techniques including Master Theorem and substitution method.

(BL-1, BL-2, BL-3)

Assessment Tool Weightage:

QUESTIONS: 10 **Total Marks: 100**

Annexure A

Concept Mapping

Code of Conduct:

I S.Chandra Siddhardha(192525222) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.

I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student:

S.Chandra Siddhardha

RUBRICS FOR CALCULATION

Criteria	**Weigh tage**	**Level 4 – Exemplary**	**Level 3 – Proficient**	**Level 2 – Developing**	**Level 1 – Beginning**
Concept Identification	25%	Identifies all key concepts from literature accurately and comprehensively	Identifies most key concepts with minor omissions	Identifies limited concepts; some are irrelevant or missing	Fails to identify essential concepts
Linkages	25%	Establishes clear, meaningful, and accurate relationships between concepts	Shows correct relationships with minor gaps	Relationships are weak, unclear, or partially incorrect	No meaningful relationships established
Organization	20%	Concepts are well-structured hierarchically with logical flow and grouping	Mostly organized with minor structural issues	Some organization but lacks clear hierarchy	No logical organization or structure
Integration of Literature	20%	Integrates multiple studies effectively to show connections and comparisons	Shows some integration of literature	Limited integration; mostly isolated concepts	No integration of literature evidence
Clarity	10%	Concept map is clear, neat, visually structured, and easy to interpret	Generally clear with minor readability issues	Clarity is reduced; difficult to interpret in parts	Poorly presented and hard to understand

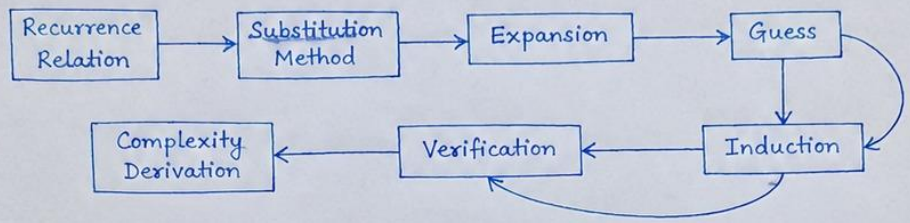
4. Consider the concept map:

Recurrence Relation → Substitution Method → Expansion → Guess → Induction → Verification → Complexity Derivation.

Trace the logical flow from a recurrence relation to its final complexity using the substitution method. Add suitable directional links to show how guessing and induction support the proof of the derived complexity. Justify the role of verification in confirming the correctness of the assumed solution. Interpret how this map provides a stepwise framework for solving recurrence relations systematically.

Answer:

The concept map with directional links is shown below:



1. Recurrence Relation → Substitution Method
   - We start with a recurrence relation, e.g., $T(n) = aT(n/b) + f(n)$.
   - The substitution method assumes a form for the solution and substitutes it into the recurrence to simplify it.
2. Substitution Method → Expansion
   - The recurrence is expanded (unfolded) using the assumed solution to see how the terms grow.
3. Expansion → Guess
   - From the expanded form, we guess the solution (e.g., $T(n) = n^k$ or $T(n) = n^k \log n$).
4. Guess → Induction
   - The guessed solution is proved correct using mathematical induction.
   - Induction shows that if the solution holds for smaller inputs, it also holds for n.
5. Induction → Verification
   - We substitute the solution back into the original recurrence and verify that it satisfies the equation.
6. Verification → Complexity Derivation
   - Once the solution is verified, we express the final result in asymptotic notation (e.g., $O(n)$, $O(n \log n)$, etc.).

Justification of Verification: Verification is essential because it confirms that the guessed solution actually satisfies the original recurrence. Without verification, the solution might be incorrect even if it looks valid after substitution and induction.

Thus, this concept map provides a clear, stepwise framework to solve recurrence relations systematically, from the initial relation to the final complexity.